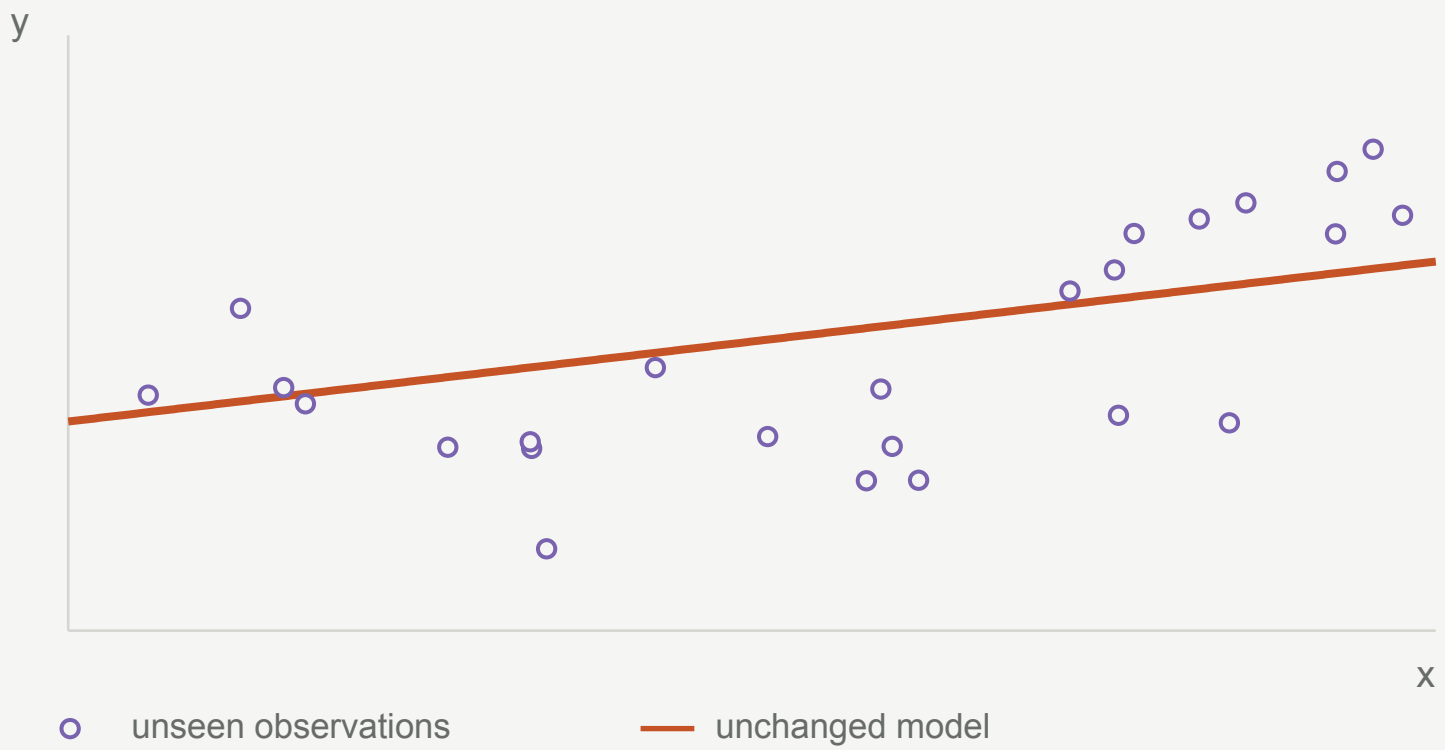
Which model would you trust?



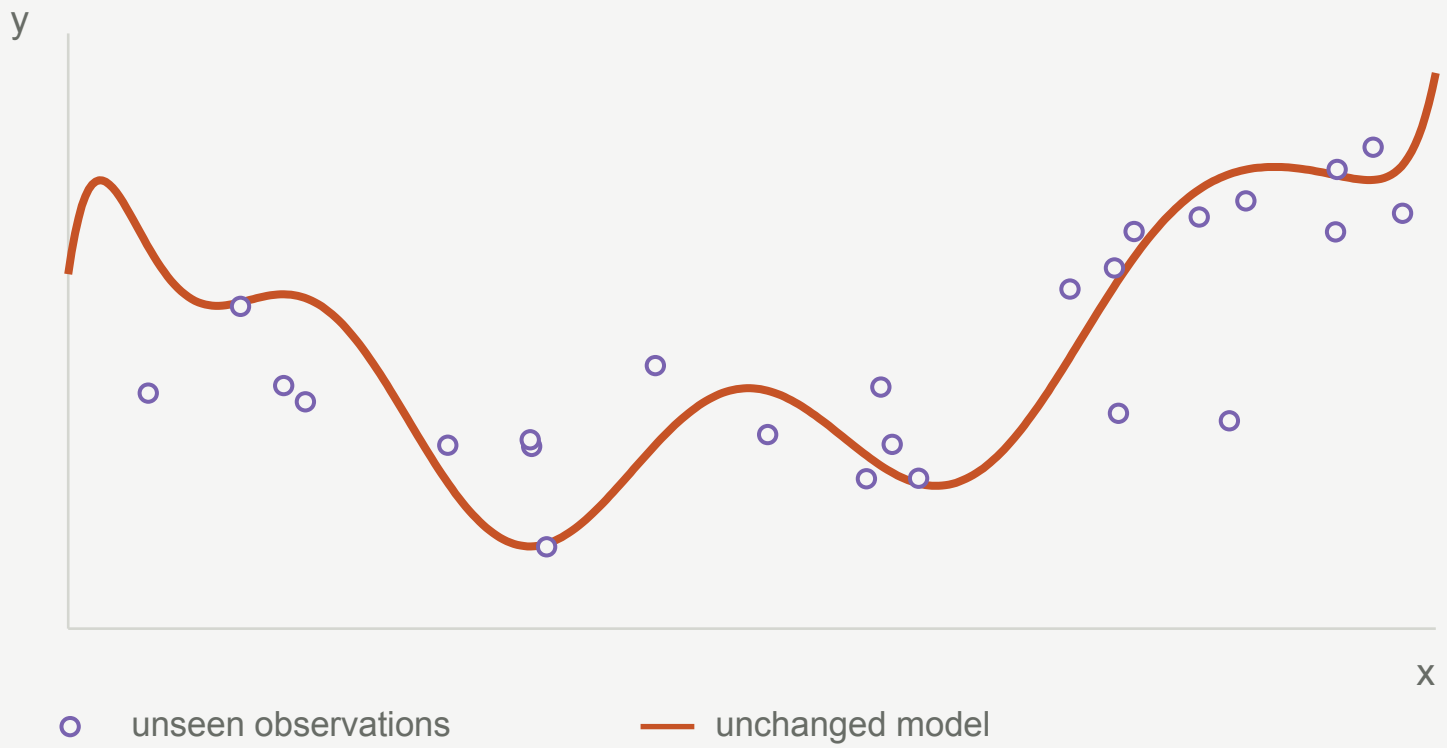
Too simple to learn the pattern.



High bias

A straight line cannot learn this curve.

Too complex to learn the pattern.



High variance

Small changes in data.
Big changes in the fit.

Change the sample. Fit again.



A closer fit is not always a better model.



Choose using validation data.

Schematic, not measured

The smallest training loss can lose.

Fit	Model degree	Training MSE ↓	Testing MSE ↓
Underfitting	1	0.444	0.336
Good fit	**2**	**0.131**	**0.170**
Overfitting	12	0.053	0.269

Learn the pattern.
Not the noise.

BIAS

Too rigid to capture the relationship.

VARIANCE

Too sensitive to the training sample.

GENERALIZATION

How well it predicts new observations.

Tune on validation. Report on the untouched test set.